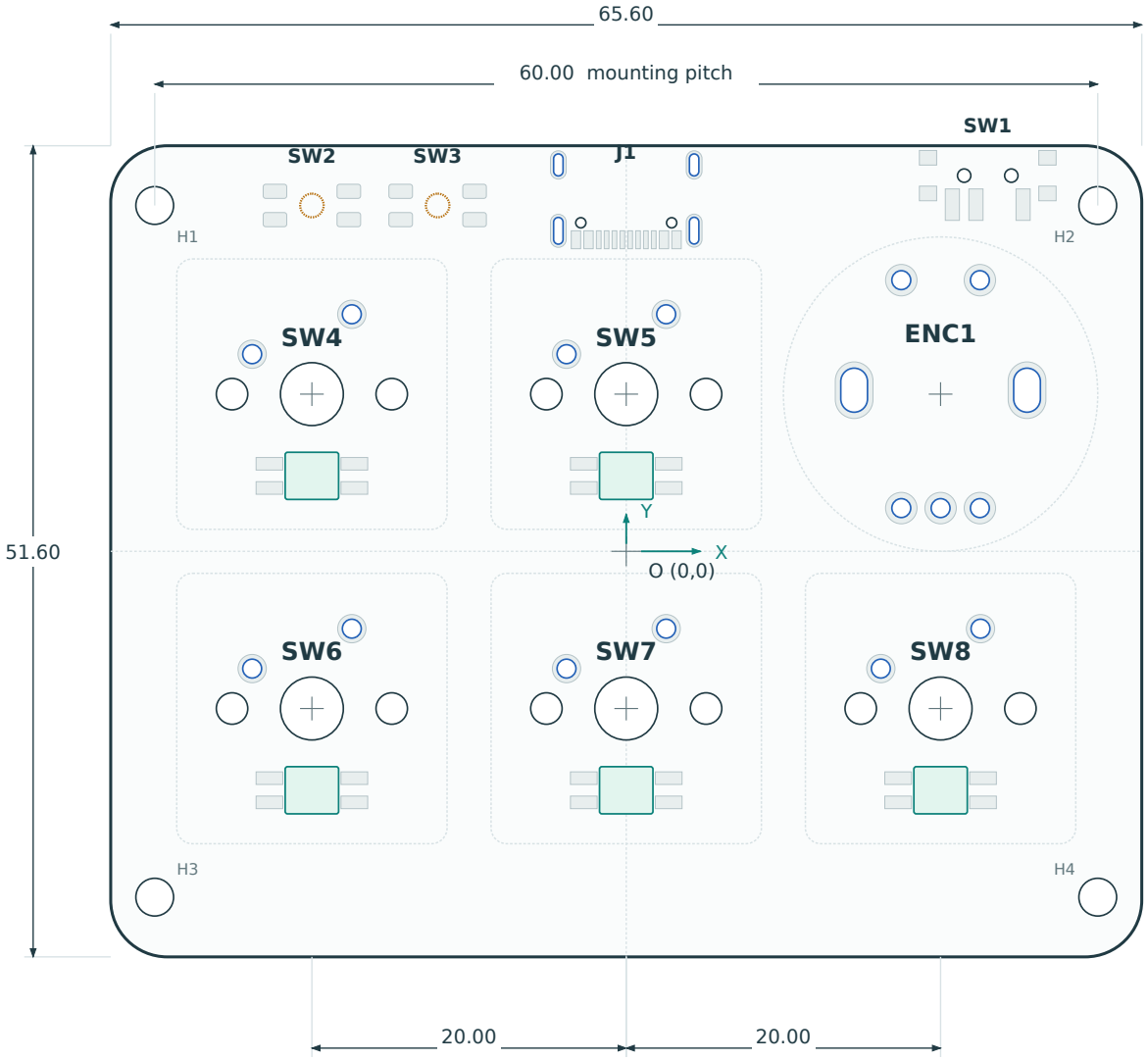
CH592 / PCB mechanical layout

Board 65.6 × 51.6 × 1.6 · R3.6 corners · 20 mm key pitch · Print04 structural proposal



FRONT / USER SIDE · dashed outlines = lid openings / knob envelope

DATUM COORDINATES

Ref.	X	Y
SW4	-20.00	+10.00
SW5	+0.00	+10.00
SW6	-20.00	-10.00
SW7	+0.00	-10.00
SW8	+20.00	-10.00
ENC1	+20.00	+10.00
J1	+0.00	+22.20
SW1	+23.00	+23.90
SW2	-20.00	+22.00
SW3	-12.00	+22.00

H1 / H2: X = ±30, Y = +22

H3 / H4: X = ±30, Y = -22

4 × Ø2.40 NPTH for M2 screws

- PCB mechanical / NPTH hole
- PCB plated hole / slot
- RGB through-board light window
- Ø1.50 lid access only (SW2 / SW3)

Board plane dimensions before 6° tilt.

Lower-left bounding-box origin:

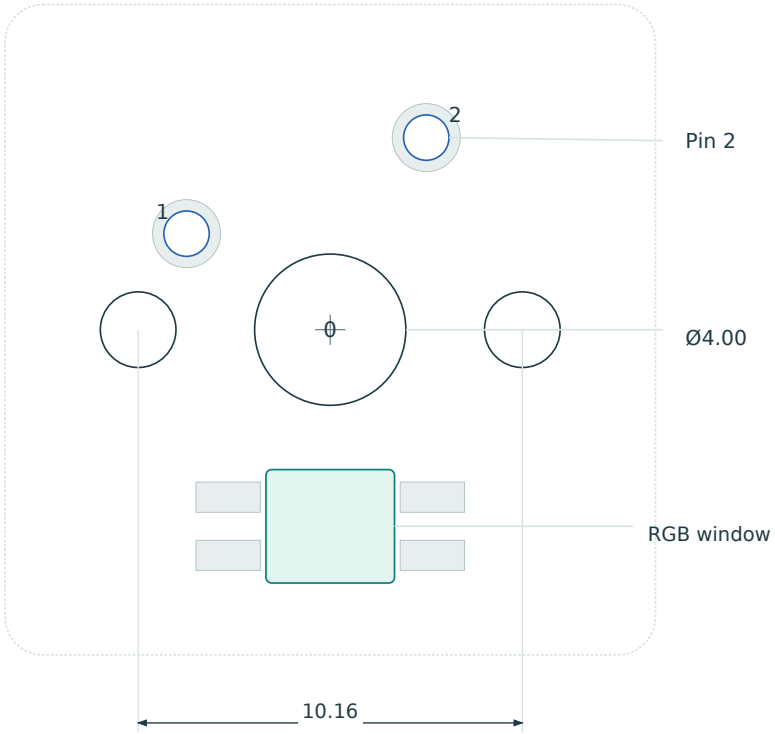
XII = X + 32.8; YII = Y + 25.8

CH592 / Key switch and encoder drill details

Local datums at switch stem / encoder shaft centers. Repeat key pattern at SW4-SW8 centers from sheet 1.

A SAMZO 28115181DM

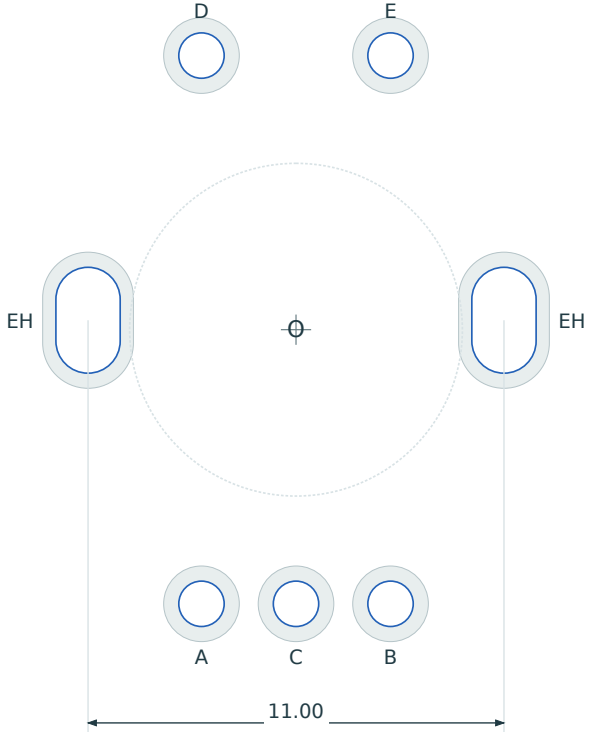
5 positions · footprint-derived pattern · detail scale 5:1



Feature	Local X	Local Y	Drill / cut
Stem locator	0.00	0.00	Ø4.00 NPTH
Side locators	±5.08	0.00	2 × Ø2.00 NPTH
Pin 1	-3.80	+2.54	Ø1.20 PTH
Pin 2	+2.54	+5.08	Ø1.20 PTH
RGB window	0.00	-5.20	3.40 × 3.00 R0.15

B EC11N1525404 / ENC1

Datum at board (20,10) · detail scale 5:1



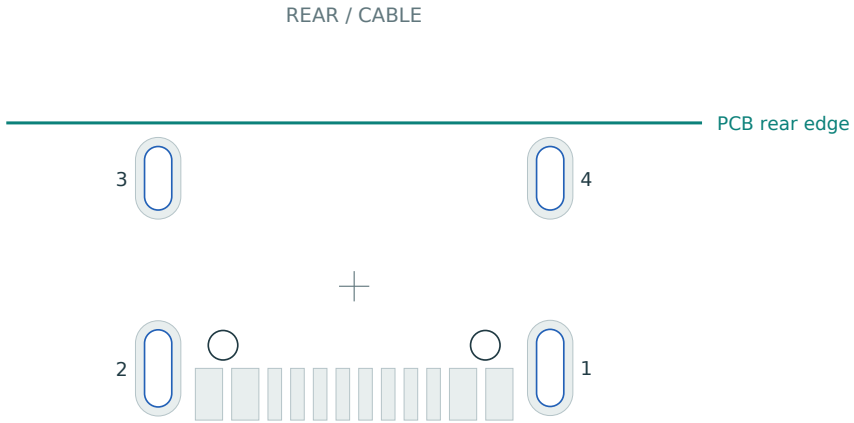
Feature	Local X	Local Y	Drill / cut
A / C / B	-2.5 / 0 / 2.5	-7.25	3 × Ø1.20 PTH
D / E	-2.5 / 2.5	+7.25	2 × Ø1.20 PTH
EH shell tabs	±5.50	+0.25	1.70 × 2.80 slot
Lid shaft bore	0.00	0.00	Ø8.80 - lid only
Hole decimals	See CSV	Source values	Nominals rounded here

CH592 / Rear interfaces and service access

USB origin: (0,22.20), 180° · Power switch BODY center: (23,23.90) · Header origins are not body datums.

C GT-USB-7010ANS / J1

Pads and holes from Rev.D · detail scale 6:1



12 SMT contacts: full numbered coordinates in CSV

Hole	Local X	Local Y	Drill
Shell 1	+4.32	-1.80	0.60 × 1.70 slot
Shell 2	-4.32	-1.81	0.60 × 1.70 slot
Shell 3 / 4	-4.32 / +4.32	+2.38	0.60 × 1.40 slot
Locators	±2.89	-1.30	2 × Ø0.65 NPTH

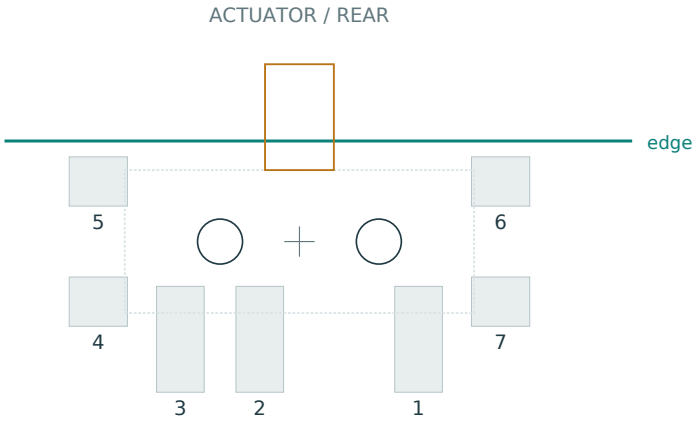
Nearest USB rear copper to PCB edge: 0.32 mm nominal.

Case throat: 12.0 × 6.0; outer mouth: 13.2 × 7.2.

Case dimensions above are openings, not PCB cuts.

D C963206 / MK-12C02-G020 / SW1

LCSC / EasyEDA footprint · body center datum · 7:1



Item	Specification
Body / travel	6.60 × 2.70 × 1.40 / 1.50
Locating pegs	2 × Ø0.85 PCB holes; X = ±1.50
Signal lands	0.90 × 2.00 (LCSC footprint)
Anchor lands	1.10 × 0.93 (LCSC footprint)

Manufacturer PDF uses smaller suggested copper lands.

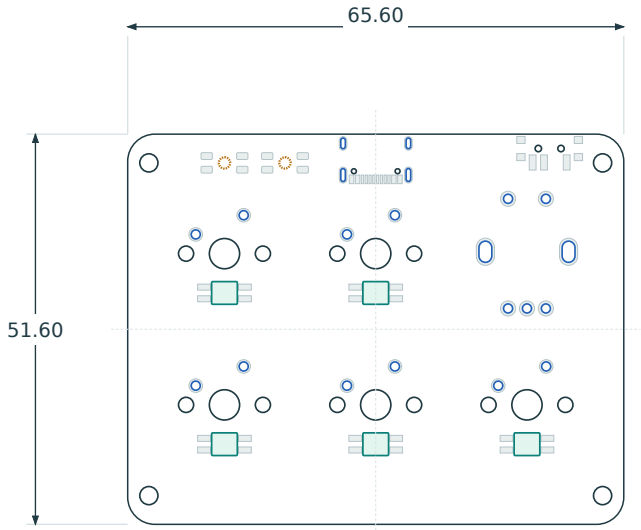
SW2 RESET (-20,22) / SW3 BOOT (-12,22)

Ø1.50 access holes in LID ONLY. Tactiles have no drills.

GT-TC029B-H025-L1N: four 1.50 × 0.90 SMT pads.

CH592 / 1:1 alignment reference

Print at Actual size / 100%. Disable Fit to page. Check the 50 mm scale bar before comparing against hardware.



FRONT / USER



50.00 mm calibration

This page is an alignment template, not a drill program.

Dashed orange circles are lid service holes only.

M2 mounting holes: Ø2.40, non-plated.

RELEASE NOTES

01 Datums

All coordinates are in the PCB plane.

X increases right; Y increases toward USB.

App transform: appx = X, appz = -Y.

02 Source fidelity

Key, encoder and USB drill patterns come from Rev.D footprints. CSV retains source decimals.

Blank mechanical holes are proposed as NPTH.

03 RGB / service access

D2-D6 are on B.Cu, mirrored in X in top view.

Confirm pin 1 and chain direction before routing.

SW2 / SW3 access holes are in the lid only.

04 Before PCB manufacture

No net assignment or copper routing is provided.

Verify real component dimensions, pad clearances, stack-up, RF placement and full DRC before release.

Files: 4 SVG sheets, PDF, coordinates CSV, mechanical DXF, data JSON and Chinese notes.